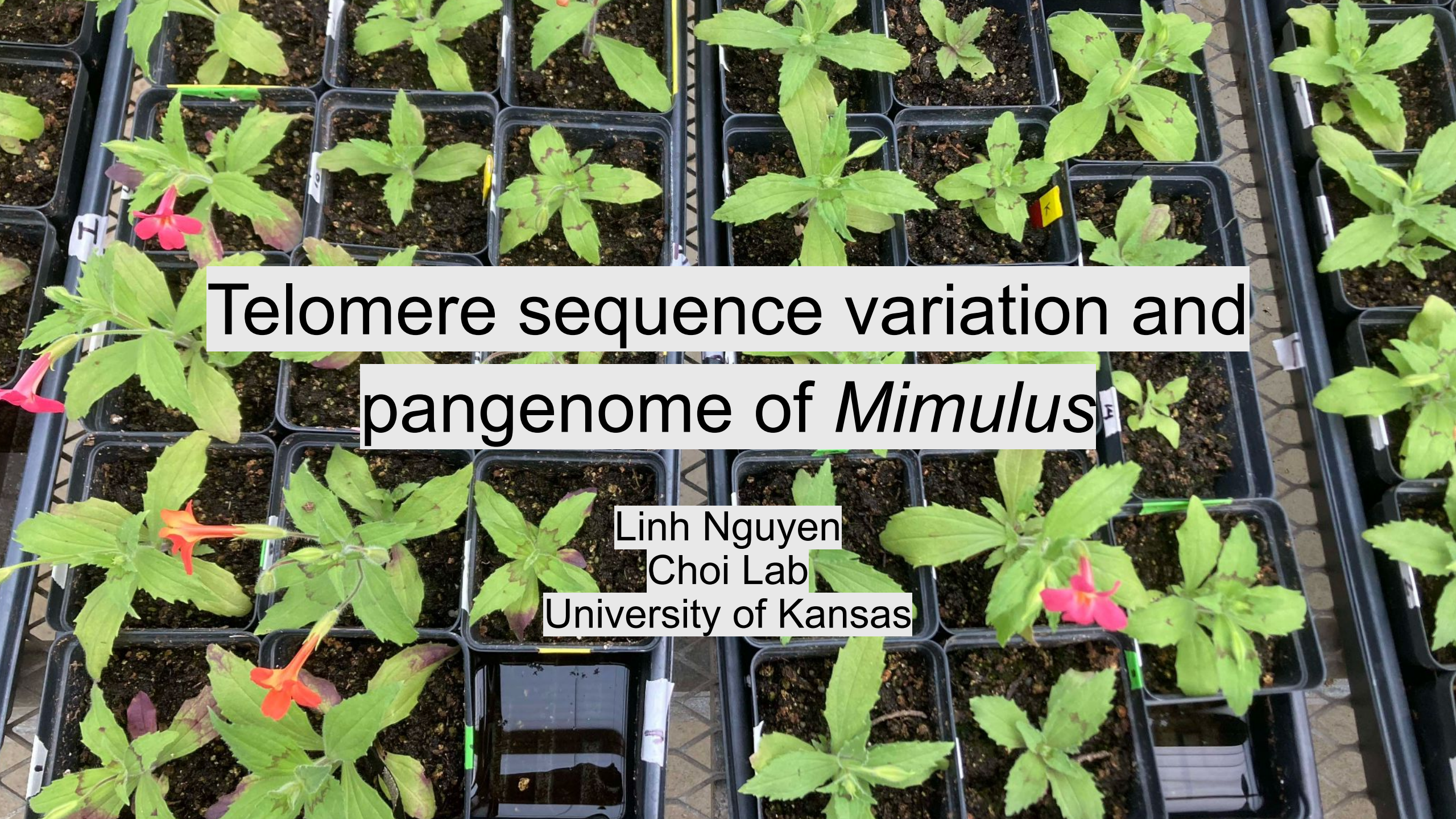




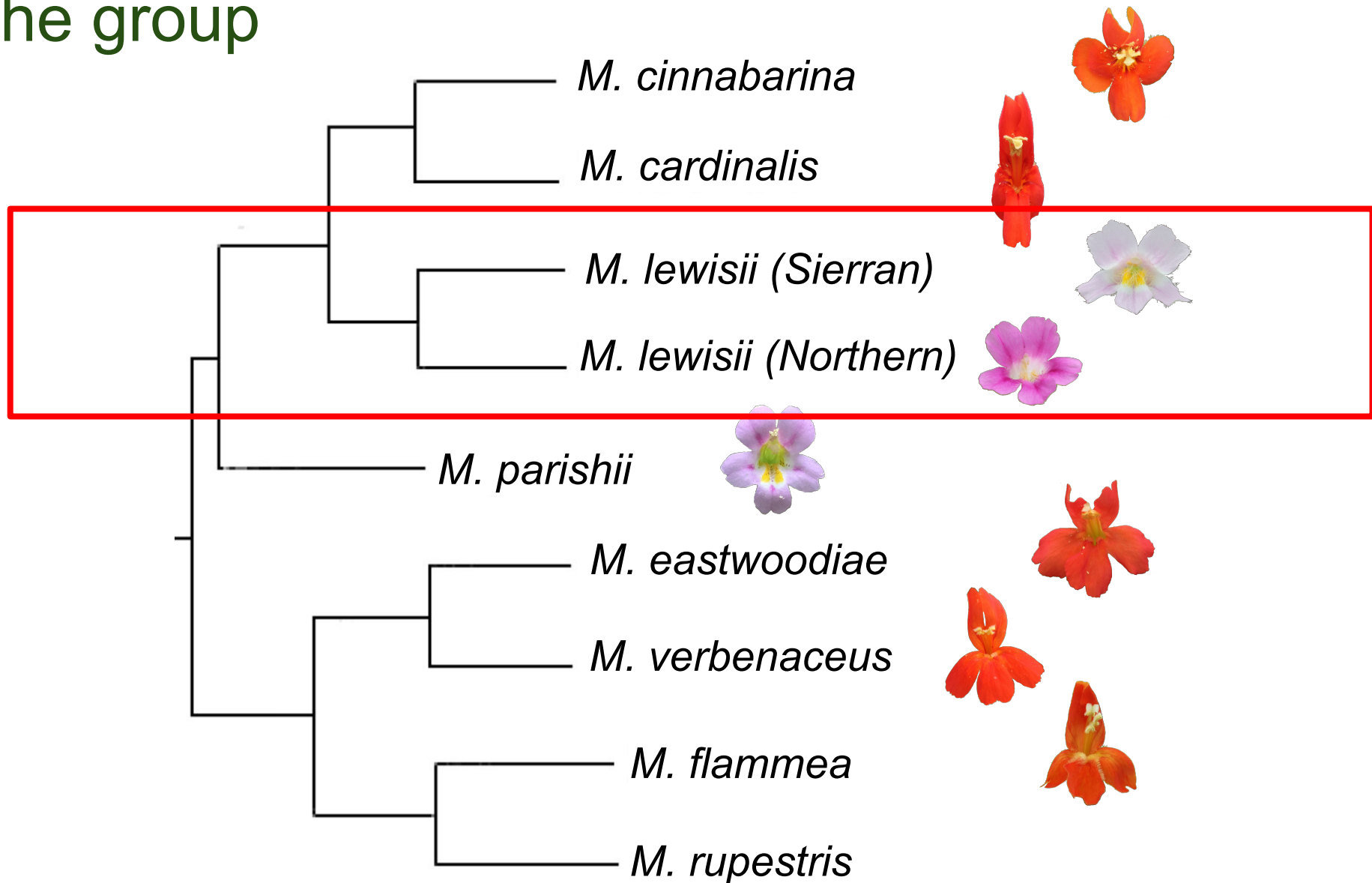
Telomere sequence variation and pangenome of *Mimulus*

Linh Nguyen
Choi Lab
University of Kansas

Variation of telomere motifs in plants



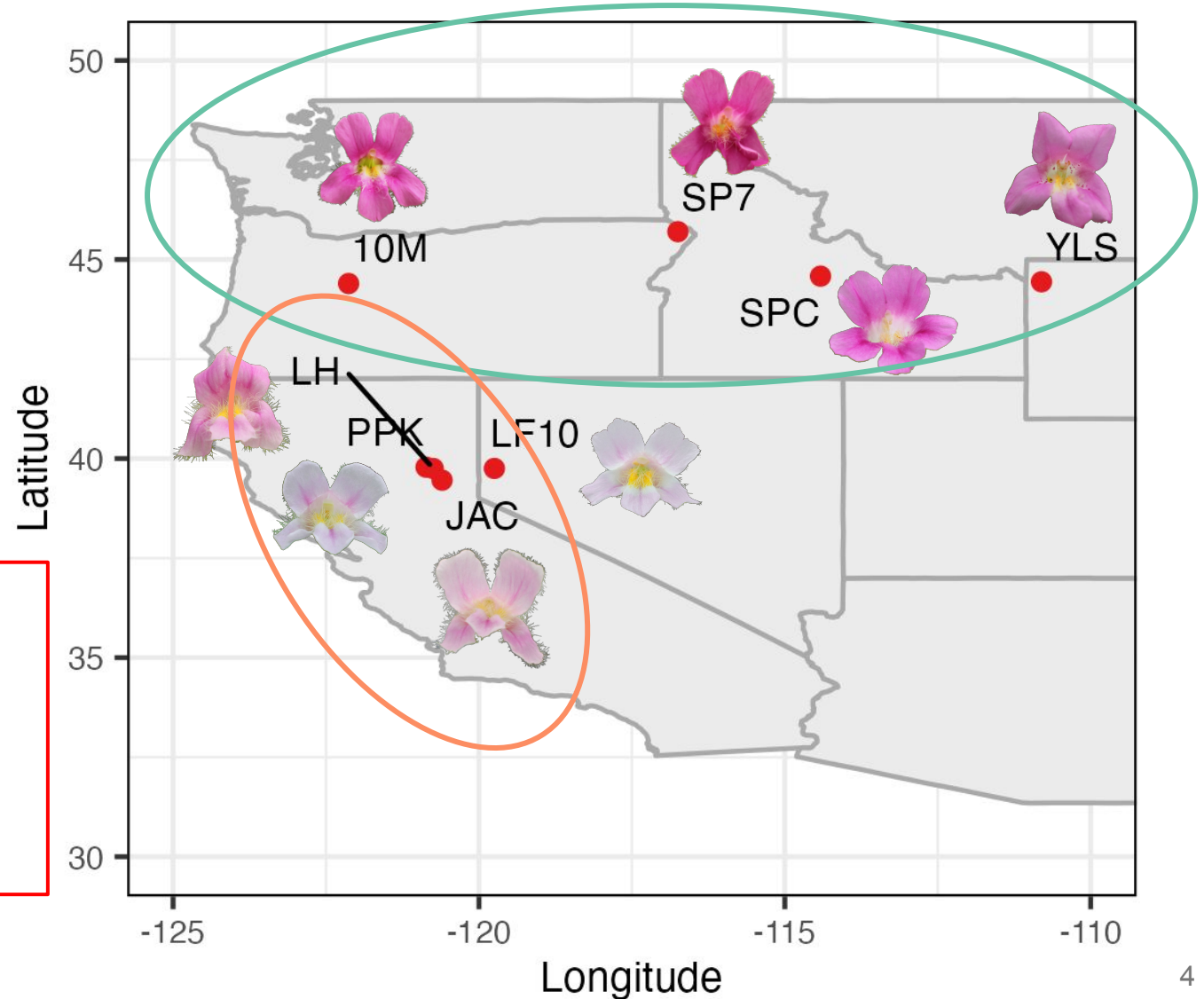
Erythranthe group



Differences between Northern and Sierran *M. lewisii*

- Geography
- Floral morphology and coloration

What are the evolutionary patterns in the telomeres for the Northern vs. Sierran population?

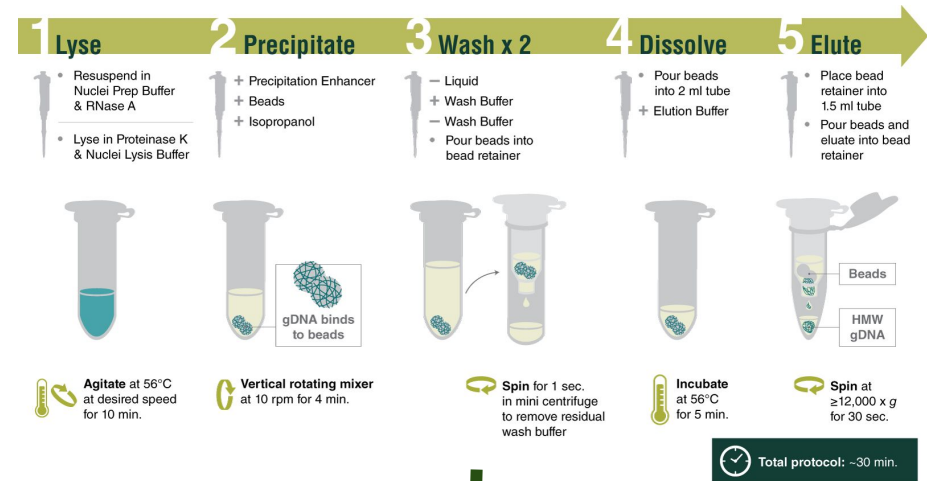


M. lewisii data collection & analysis workflow

Northern vs Sierran



High molecular weight DNA extraction



Sequences statistics +

Telomere pattern and length
quantification -
Topsicle



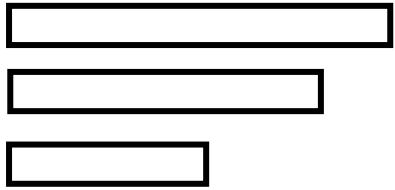
ONT sequencing + duplex basecalling



Topsicle: A method to measure telomere length from long read sequences

STEP 1: Subsetting telomere

Long read sequence



Telomere pattern

AAACCG

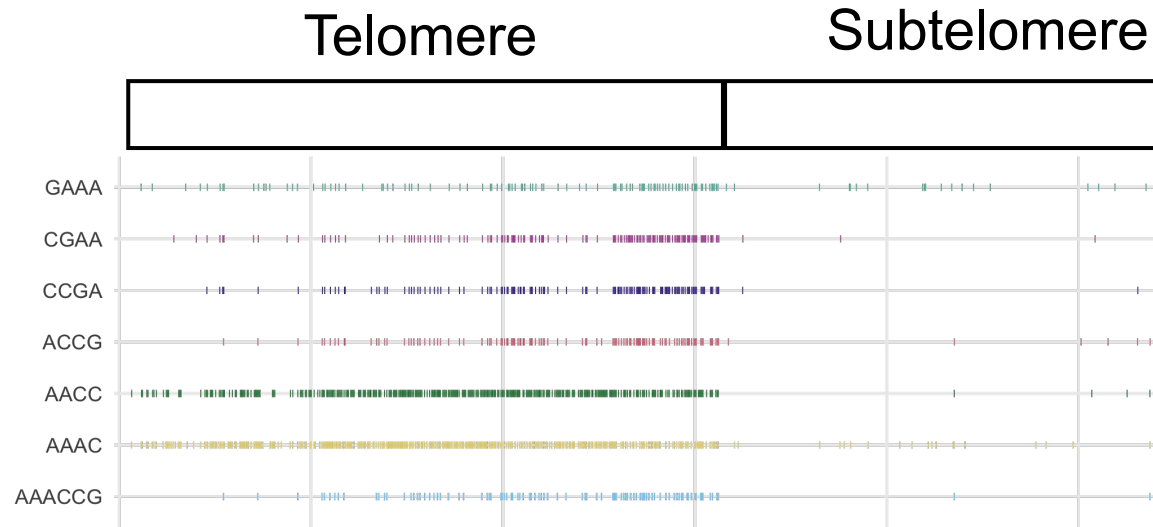
k-mer search

AAAC

AACC

ACCG

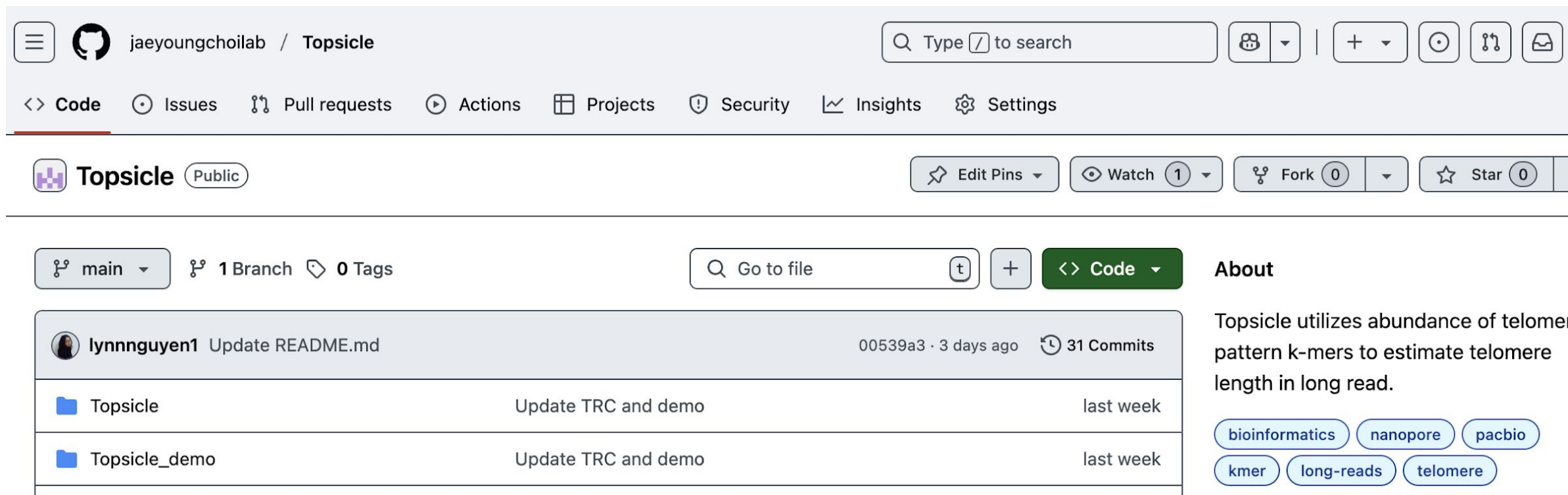
STEP 2: Quantifying k-mer distribution



STEP 3: Telomere length estimation



Topsicle is available on GitHub



The screenshot shows the GitHub repository page for **jaeyoungchoilab / Topsicle**. The repository is public and has 0 stars, 0 forks, and 1 watch. The repository description states: "Topsicle utilizes abundance of telomere pattern k-mers to estimate telomere length in long read." The repository includes two folders: **Topsicle** and **Topsicle_demo**, both updated last week. The repository is managed by **lynnnguyen1**, who updated the README.md file 3 days ago (commit 00539a3) and has made 31 commits in total. The repository is categorized under **bioinformatics**, **nanopore**, **pacbio**, **kmer**, **long-reads**, and **telomere**.

jaeyoungchoilab / **Topsicle**

Search: Type / to search

Code Issues Pull requests Actions Projects Security Insights Settings

Topsicle (Public)

Edit Pins Watch (1) Fork (0) Star (0)

main 1 Branch 0 Tags

Go to file + Code

About

Topsicle utilizes abundance of telomere pattern k-mers to estimate telomere length in long read.

bioinformatics nanopore pacbio kmer long-reads telomere

lynnnguyen1	Update README.md	00539a3 · 3 days ago	31 Commits
Topsicle	Update TRC and demo	last week	
Topsicle_demo	Update TRC and demo	last week	

github.com/jaeyoungchoilab/Topsicle

